

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is: when a client gets too long to get a response from a server, the connection is dropped.

The logs show that: a large number of TCP SYN requests coming from an unknown IP address.

This event could be TCP SYN flooding attack

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. Firstly, a SYN packet is sent through my device to the web server. It shows it wants to establish a connection.
2. Secondly, the server sends a SYN-ACK request to my device that the server acknowledges the connection. The server will reserve some resources for the final step of the handshake.
3. Lastly, an ACK request will be sent by my device to the web server to complete the TCP three-way handshake and establish the connection.

Explain what happens when a malicious actor sends a large number of SYN packets all at once: they trigger a TCP SYN Flood Attack, which is a type of Denial of Service (DoS) attack. As we know, it is a three-way handshake and by sending SYN packets to the web server it never receives ACK packets. The connection remains in a half-open state. This results in the server crash.

Explain what the logs indicate and how that affects the server: The log indicates that from the IP address "203.0.113.0" a large number of SYN packets were sent at once to the web server. As this is a TCP SYN Flood attack the server couldn't handle this request and crashed. This is why the website is giving a connection timeout error message as the server crashed it is taking time to make a connection with the server, but it cannot make the connection.